

Operating System Question Bank

Unit 5: I/O and File Management

1. List and briefly explain three techniques for performing I/O.
2. Discuss at least two techniques for performing I/O.
3. Explain in brief different I/O buffering techniques.
4. Enlist the characteristics of block and character devices? Explain each with suitable example.
5. Define the following terms:
 - a. Seek Time
 - b. Rotational Latency
 - c. Transfer Time
6. What are the main functions of disk management?
7. Define the following terms: Seek time, Rotational Latency and Transfer Time.
8. Assume the disk head is initially positioned over track 100. For the disk track request 27, 12, 110, 186, 147, 41, 10, 64, 120 how disk scheduling is done for FIFO Scan algorithm. Calculate average seek length and show the tracing of request.
9. Explain Disk Structure.
10. Assume a disk with 200 tracks and the disk request queue has random requests in it as follows: 55, 58, 39, 18, 90, 160, 150, 38, 184. Find the no. of tracks traversed and average seek length if SSTF, SCAN and C-SCAN algorithms are used and initially head is at track no 100.
11. For the given sequence of disk request, determine the total distance travelled by disk head in satisfying the entire request for FCFS, C-SCAN and SSTF algorithms. Initial head position is 120 and total number of cylinders in the disk is 200.
120, 130, 180, 150, 25, 10 105, 90
12. Assume a disk with 200 tracks and the disk request queue has random requests in it as follows: 98, 183, 37, 122, 14, 124, 65, 67. Find the no. of tracks traversed and average seek length if FCFS, SSTF, SCAN and C-SCAN is used and initially head is at track no 53.
13. Is disk scheduling, other than FCFS, useful in a single user environment. Explain your answer.
14. What are the different disk performance parameters? Which scheduling algorithm gives best performance.
15. Suppose that a disk drive has 400 cylinders, numbered 0 to 399. The drive is currently serving a request at cylinder 160. The queue of pending requests, in FIFO order is 370, 30, 390, 130, 310, 170, 340, 180. Starting from the current position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending request, for each of the following disk scheduling algorithms?
 - i) FCFS
 - ii) SSTF

iii) SCAN

iv) LOOK

16. Describe different types of disk scheduling algorithms?
17. Consider a disk with N tracks numbered from 0 to $(N-1)$ and assume that requested sectors are distributed randomly and evenly over the disk. We want to calculate the average number of tracks traversed by a seek. Calculate the probability of a seek of length j when the head is currently positioned over track t .
18. Why we need disk scheduling?
19. Compare SCAN and C-SCAN scheduling algorithms with example.
20. Differentiate between LOOK and C-LOOK disk scheduling algorithm?
21. What are the functions of a file management system?
22. Explain the various file attributes. Also explain the various file operations.
23. List and briefly define the different file types.
24. Explain the concept of directory structure.
25. Explain the different types of access methods.
26. Explain the different file system performance issues.
27. Difference between sequential file and index file.
28. Explain the concept of file directories with the help of example.
29. What is single-level and two-level directory with the help of an example?
30. What are typical operations that may be performed on a directory?
31. List the different types of directory structure and briefly explain them.